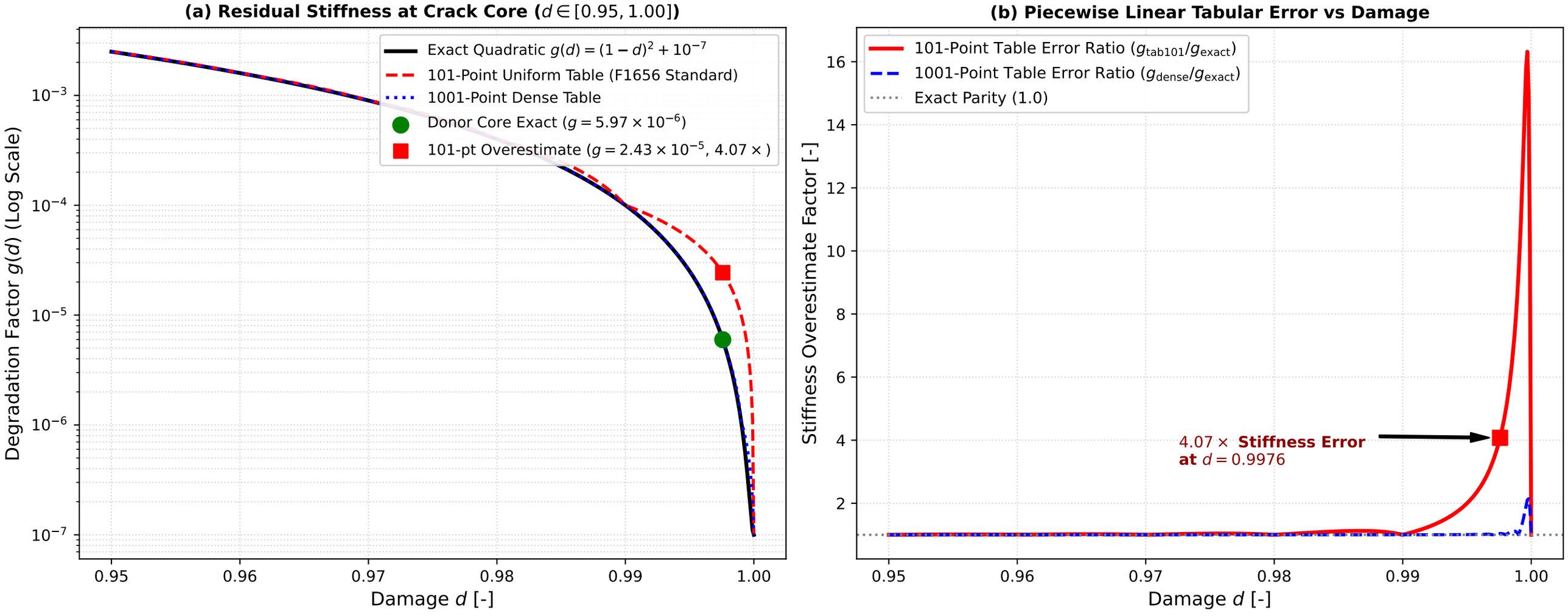
(a) Residual Stiffness at Crack Core ($d \in [0.95, 1.00]$)
Exact Quadratic $g(d) = (1-d)^2 + 10^{-7}$
101-Point Uniform Table (F1656 Standard)
1001-Point Dense Table
Donor Core Exact ($g = 5.97 \times 10^{-6}$)
101-pt Overestimate ($g = 2.43 \times 10^{-5}$, $4.07 \times$)
Degradation Factor $g(d)$ (Log Scale)
10^{-3}
10^{-4}
10^{-5}
10^{-6}
10^{-7}
0.95
0.96
0.97
0.98
0.99
1.00
Damage d [-]
(b) Piecewise Linear Tabular Error vs Damage
101-Point Table Error Ratio (g_{tab101}/g_{exact})
1001-Point Table Error Ratio (g_{dense}/g_{exact})
Exact Parity (1.0)
Stiffness Overestimate Factor [-]
16
14
12
10
8
6
4
2
4.07 × Stiffness Error
at $d = 0.9976$
0.95
0.96
0.97
0.98
0.99
1.00
Damage d [-]